Task Allocation Sheet

Formative 2: Deep Q-Network (DQN) on Atari Boxing · Group repository: github.com/Teddy-2004/dqn-atari

Environment	`ALE/Boxing-v5` (Gymnasium / ALE)
Group members	Tedla Tesfaye Godebo · Parfait Uhiriwe · Hasbiyallah Umutoniwabo
Tools	PyTorch, Gymnasium, uv (package manager), GitHub

1. Task allocation

#	Task	Owner	Deliverable	Status
1	Environment selection and project setup (uv)	Tedla	`pyproject.toml`, `uv.lock`, choice of Boxing	**Done**
2	DQN implementation: Q-network (CNN), replay buffer, target network, ε-greedy, Bellman update	Tedla	`train.py`, `common.py`	**Done**
3	Evaluation script (greedy play, real-time display)	Tedla	`play.py`	**Done**
4	Experiment comparison tool and group instructions	Tedla	`compare_experiments.py`, group setup and experiment guide	**Done**
5	10 hyperparameter experiments: learning rate, gamma, batch size, ε schedule (baseline sweep)	Tedla	`models/Teddy/`, `Teddy_Hyperparameter_experiment_results.md`	**Done**
6	10 hyperparameter experiments: learning rate and batch size	Parfait	`models/Parfait/`, `Parfait_Hyperparameter_experiment_results.md`	**Done**
7	10 hyperparameter experiments: gamma and exploration (ε start, end, decay)	Hasbiyallah	`models/Hasbiyallah/`, `Hasbiyallah_Hyperparameter_experiment_results.md`	**Done**
8	Final model selection (greedy test of the top 4 models over 10 rounds)	Tedla	`dqn_model.pth` (Hasbiyallah's exp 10), README evaluation table	**Done**

9	README: introduction, architecture diagrams, training process, results	Parfait	`README.md`	**Done**
10	Gameplay demonstration video	Parfait	YouTube video linked in the README	**Done**
11	Task allocation sheet	Tedla	`Task allocation sheet.pdf`	**Done**
12	Presentation slot booking (Week 6)	Hasbiyallah	Booked slot with the coach	**To confirm**
13	Presentation slides	Hasbiyallah	Slide deck (one section per presentation part)	**To confirm**
14	Live demo of `play.py`	Tedla	Live run during the presentation	**On the day**

2. Presentation roles (10 minutes)

Part	Time	Presenter	Content
Introduction	1 min	Parfait	Boxing environment, goal of the game, scoring
DQN implementation	2 min	Tedla	Q-network, replay buffer, target network, ε-greedy, Bellman equation and loss
Hyperparameter experiments	2 min each	Tedla, Parfait, Hasbiyallah	10 experiments each: best and worst runs, what helped or hurt, final choice
Live demo	1 min	Tedla	Run `play.py` with the final model
Q&A	10 min	All members	Every member answers questions

Statuses reflect the repository as of 11 November 2026.